



Gender-related aspects of invention networks: A firm-level analysis

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ABSTRACT

This paper integrates insights from the literature on invention networks, gender, and the sociological literature to analyze differences in how firms participate in man-led and woman-led invention networks. We contribute to the current debate on whether clustering or boundary-spanning network properties are more important for invention by introducing gender as an important factor. We empirically test our hypotheses on a sample of more than 30,000 firms from around the world over time using OECD REGPAT global patent data. Our findings indicate that different network properties are important for firm invention in woman-led and man-led innovation networks. In man-led invention networks, firms strongly benefit from being in a boundary-spanning position and are negatively affected by clustering, whereas in woman-led invention networks, boundary spanning has a less pronounced positive effect, and clustering has a positive rather than negative effect. Our findings have substantial implications for firms and policymakers interested in invention and contribute to the studies of gender and invention networks.

1. Introduction

In today's highly competitive economy, firms' survival and performance increasingly rely on their ability to produce patented inventions (Hagedoorn et al., 2003). Patents provide strategic, financial, and competitive advantages (Cantrell, 2009). While much research focuses on firms' financial, technological, and policy contexts, a growing body of work highlights the role of social structures—especially internal social organization and networks—in shaping invention performance (Guler and Nerkar, 2012; Khanna and Guler, 2022). Additionally, firms are no longer self-sufficient in invention, relying instead on inter-firm collaborations and knowledge sharing through co-invention (Sytych and Tatarynowicz, 2014; Liebeskind et al., 1996).

Social network theory highlights a firm's external network as a key driver of invention performance, as it facilitates access to complementary resources through partnerships (Audretsch and Belits, 2023). For instance, Calia et al. (2007) showed that participation in an external technological invention network enabled a firm in their case study to transform its business model and enhance global competitiveness. More recently, co-invention has gained traction as a strategic approach to innovation, with studies indicating that approximately one-third of firm inventions now involve external partners (Toth and Lengyel, 2017; Walsh et al., 2016).

Research on invention networks has identified two key structural positions that influence invention performance: boundary spanning (acting as a broker between unconnected actors) and clustering (forming cohesive ties with partners who are also interconnected) (Toth and Lengyel, 2021). More broadly, network literature emphasizes clustering and centrality—of which brokering is a type—as central structural positions within a network (Coleman, 1990). Clustering promotes cohesion, builds trust, and supports resource sharing (Coleman, 1990), while boundary spanning allows firms to bridge disconnected groups, enabling access to novel resources and information (Burt, 1992). Burt (2000, p. 11) argued that "... bridges through structural holes are the source of the ideas of the new inventions ...". However, the evidence remains mixed regarding the impact of key network positions on invention performance. For example, Toth and Lengyel (2021) found that both boundary spanning and clustering positively influenced firms' invention outcomes. In contrast, Le Gallo and Plunket (2020) reported that while brokering enhanced invention performance, clustering—particularly among "stars" with dense local connections—had little to no effect, or even a negative one, on invention quality. Several studies highlight the potential drawbacks of overly dense networks, suggesting that such environments may lead to complacency and isomorphism over time, where actors' ideas and practices become too similar due to a lack of external knowledge (Bathelt et al., 2004). Conversely, Long et al. (2013)

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showed that densely connected networks can more efficiently disseminate information among members than sparsely connected ones, suggesting that while boundary spanning is valuable, it may not always be the most efficient mode of information transfer, because brokers—who often serve as gatekeepers of specialized knowledge or resources—can become overwhelmed by their responsibilities and may require additional support to function effectively.

In this paper, we propose that these inconsistencies may stem from the literature's limited consideration of gender in inter-firm invention networks. Specifically, we suggest that the effects of boundary spanning and clustering may differ between woman-led and man-led innovation networks. More broadly, existing research on invention networking tends to overlook the gender profiles of inventors and how gender-based differences may shape networking strategies and, in turn, influence firms' invention performance. To quote [Arun and Rojers \(2021\)](#), "in the absence of such research, it is challenging to formulate public and firm-level policies to encourage innovation." Drawing on insights from sociological literature, as well as research in network and gender studies, we argue that male inventors are generally more inclined toward brokerage—that is, long-range networking strategies aimed at controlling information flows, managing risk and uncertainty, and accessing knowledge from distant sources. In contrast, female inventors may be less comfortable occupying the power-centric roles often associated with brokerage and are more likely to rely on trust and reciprocity fostered through clustered network structures.

Research on gender and firm-level innovation highlights the importance of women's participation in the innovation process, as they bring distinct perspectives, skills, and cognitive approaches ([Turner, 2009](#); [Alves et al., 2017](#); [Arun and Rojers, 2021](#)). Studies show that woman-led innovations often differ from those led by men, tending to focus more on local needs and social impact ([Nählinder, 2010](#); [Blake and Hanson, 2005](#)). For example, [Koning et al. \(2021\)](#) found that women were more likely to address gaps in areas like women's health. Women's involvement is also linked to greater interdisciplinarity in patents, as indicated by a higher number of IPC codes ([Sugimoto et al., 2015](#)). Moreover, their participation in R&D enhances not only the quantity but also the radicalness of innovation ([Diaz-Garcia et al., 2013](#); [Garcia Martinez et al., 2017](#)). Overall, women inventors make complementary contributions that expand a firm's innovation scope. At the same time, research shows that women participate less than men in patenting and innovation networks ([Tahmooresnejad and Turkina, 2022](#)), face greater challenges in securing patents, and receive fewer citations for their innovations ([Ding et al., 2006](#); [Jensen et al., 2018](#); [Rosser, 2009](#); [Ceci and Williams, 2011](#); [Legewie and DiPrete, 2014](#); [Larivière et al., 2013](#); [Pololi et al., 2013](#); [Jolly et al., 2014](#)). Gender-based inequalities also persist in the social interactions underlying innovation networks ([Brands et al., 2022](#)). Therefore, it is important to examine how firms can increase women's participation in inter-firm invention networks and which network positions support woman-led innovation.

This paper addresses these questions by integrating literature on gender and innovation, gender and networks, and sociological theory to develop a framework for understanding how boundary spanning and clustering affect man-led and woman-led networks differently. It also examines which network positions are most conducive to fostering woman-led innovation. We empirically test our hypotheses using global OECD REGPAT patent data.

Our findings indicate that in man-led networks, firms benefit significantly from occupying boundary-spanning positions and tend to be negatively impacted by clustering. In contrast, within woman-led networks, the positive effect of boundary spanning is more modest, while clustering appears to have a beneficial rather than detrimental influence.

The paper proceeds with a literature review and hypotheses, followed by data, methodology, and empirical analysis, and concludes with a discussion of findings and final remarks.

2. How gender and innovation network characteristics affect firms' invention performance

Collaborative relationships have been found to have an important impact on innovation activity ([Turkina and Van Assche, 2018](#); [Turkina and Oreshkin, 2021](#)), and collaboration among inventors from different firms is on the rise ([Whittington, 2018](#); [Toth and Lengyel, 2017](#); [Walsh et al., 2016](#)). The literature on inter-firm invention networks demonstrates that a firm's position in an invention network has important implications for its invention performance ([Ozman, 2009](#)). The principal debate in this research is structured around two network positions: boundary spanning, when a firm has long-reach connections and is a broker for many disconnected pairs of firms ([Toth and Lengyel, 2017](#)), and clustering, when a firm's partners are also connected to one another ([Boschma et al., 2015](#)). One does not necessarily exclude the other: a firm can have high clustering with some partners and, at the same time, a high number of brokering connections with others ([Barabasi, 2018](#)). Nevertheless, existing studies show inconsistent evidence with regard to how these network positions impact invention performance. Some studies argue that both brokering and clustering have a positive effect on invention: a firm in a brokering position controls the flow of information and is able to obtain knowledge from different sources and to recombine it to produce new innovations, while clustering allows to spread new ideas and knowledge faster into the local milieu ([Toth and Lengyel, 2017](#)). Other studies argue that clustering has a negative effect because all cluster members have access to the same pool of information, which may create a lock-in effect and lead to isomorphism ([Li et al., 2013](#)). Some scholars argue that, on the contrary, clustering is more beneficial for innovation than brokering is because it facilitates social capital and promotes trust among cluster members, which leads to a productive work culture ([Karlsson and Stough, 2005](#)). We posit that these inconsistencies in the literature can be reconciled by considering the gender profile of inventors as a differentiating factor for invention performance. In the following subsections, we explore how woman-led and man-led innovation networks are different and how boundary spanning and clustering have different effects in these networks. To the best of our knowledge, no study has yet looked into how firms' participation in man-led and woman-led innovation networks differs or network-related factors that specifically facilitate woman-led innovation. Only two studies ([Whittington, 2018](#); [Tahmooresnejad and Turkina, 2022](#)) have focused on gender-related aspects of innovation networks, and both consider the individual inventor level (not the firm level) and show that (a) women's patenting activity increases over time and (b) women's participation in inventor teams contributes significantly to patent productivity and quality.

However, inventors do not operate in a vacuum; they work in corporate structures that often define the extent and limits of collaborative arrangements. We do not yet know which firms are the most active in global woman-led innovation networks, whether encouraging women inventors to lead patenting teams and network with inventors from other firms has an impact on firms' innovation productivity and quality, or which network positions influence woman-led innovation. In the following subsections, we combine concepts from the literature on gender and social networks and the literature on gender and innovation with insights from sociological studies to develop a comprehensive framework for analyzing how a firm's network position influences its invention performance from a gender-based perspective.

2.1. Firm-level boundary spanning and invention performance

Several studies on inter-firm networks have demonstrated that brokerage plays an important role in enhancing firms' invention performance ([Woods et al., 2022](#); [Toth and Lengyel, 2017](#)). Being a broker enables a firm to have access to a greater abundance and variety of knowledge, recombine the knowledge and ideas it receives from other firms to a greater extent, and have more control over the flow of

information (Faulk et al., 2017; Orlic et al., 2018). It has been argued that firms with boundary-spanning connections can source knowledge, ideas, information, and expertise from their partners to produce new innovations (Uzzi and Spiro, 2005; Fleming et al., 2007). Therefore, firms whose inventors have long-range connections with inventors from other firms see an extra boost in their invention performance (Toth and Lengyel, 2017). Such inventor-brokers that connect their firm with other firms act as information brokers for both internal and external sources and thus abolish communication barriers within a company or technology field (Goetze, 2010). However, the existing literature makes no distinction between man-led and woman-led innovation networks when investigating firm brokerage.

While boundary-spanning linkages are beneficial for both men and women inventors, men inventors benefit more from them. The sociological literature indicates that men generally network more strategically and use their networks more efficiently to improve their performance and advance their career (Szell and Thurner, 2013). Men inventors are more inclined to use long-reach networking to mitigate risk and uncertainty and obtain knowledge from foreign environments (Nählinder, 2010). Moreover, existing research indicates that women inventors tend to produce different types of innovations than men inventors: woman-led innovations tend to address needs that are specific to their local environments, whereas men's innovations are more general and therefore require knowledge and ideas sourced from more distant connections (Nählinder, 2010; Blake and Hanson, 2005).

Moreover, the literature on gender and social networks that explores career and friendship networks demonstrates that women are less likely to be brokers in both instrumental and expressive networks (Fang et al., 2021). Additionally, Brands and Mehra (2019) contend that when women perceive themselves as brokers in networks, they feel threatened due to negative stereotypes about women brokers, which undermines their performance. According to researchers, women are uncomfortable in positions of power and control over the flow of information and experience anxiety in brokerage positions, which undermines their performance in these types of roles. Men, in contrast, do not experience anxiety when they perceive themselves to be brokers (Brands and Mehra, 2019).

As a result, we can expect that a firm's innovation performance in boundary-spanning positions will be driven by man-led innovation networks.

Therefore, we propose the following hypotheses:

Hypothesis 1. There is a positive correlation between a firm being in a boundary-spanning position in man-led innovation networks and its innovation performance.

Hypothesis 2. There is no correlation between a firm being in a boundary-spanning position in woman-led innovation networks and its innovation performance.

2.2. Firm-level clustering and invention performance

There is mixed evidence regarding how firm-level clustering affects firms' invention performance. For instance, Kim et al. (2020) demonstrated that firms that have tight collaborative connections among inventors have a higher probability of developing new technologies. In a similar vein, Tahmooresnejad and Beaudry (2018) found clustering has important effects on invention performance in universities: researchers who have tighter collaborations with other researchers from their local environment experience a boost in their invention performance. The literature on industrial clustering highlights the positive effects that local social networking and clustering have on invention performance (Bathelt and Li, 2014; Turkina and Oreshkin, 2021).

However, many studies have also found clustering negatively affects invention performance. For instance, a study by Tahmooresnejad and Beaudry (2019) on Canada's innovation network showed that tight clustering results in declining returns. Similarly, Beaudry and Kananian

(2013) found clustering has a negative impact on both the quantity and the quality of nanotechnology and biotechnology patents due to lock-in effects as innovations become isomorphic, which suggests a negative effect associated with too much redundancy.

We argue that these inconsistencies in the literature exist because gender-based distinctions have not yet been considered in network studies, as clustering yields different effects for firms in man-led and woman-led innovation networks. Unlike men inventors, women inventors usually tend to work in collaborative groups that are more cohesive, meaning their co-authors are also linked (Szell and Thurner, 2013). Tight collaboration has been found to be crucial for women, who are underrepresented in commercial patenting and generally less central in collaboration networks (Wittington, 2019; Tahmooresnejad and Turkina, 2023). Clustering is important for women inventors, as they are more risk averse and more dependent on social support and trust (Ibarra, 1992). Existing studies have also highlighted the fact that women inventors are more productive in cohesive collaborative groups than men inventors and tend to repeat their past collaborations (Ductor et al., 2023; Lindenlaub and Prummer, 2021). Women have also been found to feel more comfortable in groups where there is a high level of trust and in which group members are highly clustered (Szell and Thurner, 2013). While studies from other industries also generally show clustering positively affects women, a study by Lutter (2015) on the film industry found that women actors have a high risk of career failure when they are part of cohesive networks and better chances of survival when they are embedded in open, diverse structures. At the same time, the film industry is very specific and differs from the innovation domain in that actors are heavily dependent on informal networks for their roles and women in particular suffer from labor markets being structured by informal recruitment practices. This is true for the film and television industries (Grugulis and Stoyanova, 2012) and other creative industries (Christopherson, 2009). In contrast, in the field of science, recruitment depends more on formal structures, and inter-firm collaboration on patents is governed by innovation-driven pragmatic logic rather than informal friendship or acquaintance logic (Beck et al., 2022). Therefore, women inventors do not experience the same kind of closure from clustering in the innovation domain as women actors do in the film industry.

The more general sociological literature argues that women seek more social support than men in social relationships and are more productive in environments where there is a high level of social support and social capital, while men seek more efficiency, results, and new opportunities from their connections (Bourdieu and Wacquant, 1992). Burt (2000) argued that women perform better in interconnected networks and do not profit from brokerage as men do. Moreover, research on gender and networks indicates that women's connections decay faster than men's, and women lose social capital at a greater rate than men (Brands et al., 2022). Therefore, tight clustering helps women to protect their connections from decay and ensure they have substitutes for disappearing connections.

As a result, we can expect that strong clustering (a firm's location in a cohesive cluster of highly linked inventors) in man-led innovation networks may negatively affect a firm's invention performance, while clustering in woman-led innovation networks will positively impact its invention performance.

Therefore, we propose the following hypotheses:

Hypothesis 3. There is a negative correlation between a firm's level of clustering in man-led innovation networks and its invention performance.

Hypothesis 4. There is a positive association between a firm's level of clustering in woman-led innovation networks and its invention performance.

3. Data and methodology

This study leverages patent data sourced from the OECD REGPAT (Organization for Economic Co-operation and Development Regional Patent) database, which contains records of patents granted between 1976 and 2019. The database provides details about the patents, the inventors involved, and the inventors' affiliated firms. We utilized multiple gender databases, the main one being the WIPO (World Intellectual Property Organization) gender database, to accurately assess the gender of inventors. This database was essential for ensuring precise gender identification, which is a critical factor in constructing gender-specific innovation networks. It includes a comprehensive gender lexicon derived from PCT (Patent Cooperation Treaty) applications that encompasses over 6.2 million names from 182 countries. This database enabled us to reliably determine the gender of inventors based on their names and thereby facilitated the accurate categorization of first authors as men or women for our network analysis.

We developed two distinct gender-specific firm-level innovation networks by mapping the connections between inventors from each firm based on our data. The construction of these networks is predicated on the assumption that the connections between firms effectively mirror the collaborative interactions of the firms' inventors (Liu et al., 2021). The first network comprises linkages based on patents led by men inventors from 33,636 firms and thereby captures the extensive range of contributions that the men inventors who led patenting teams made to technological innovation. The second network, on the other hand, comprises linkages based on patents led by women inventors from 4249 firms. This network is designed to highlight the collaborative efforts that were spearheaded by women and the unique contributions that the women inventors who led patenting teams made to technological innovation. By isolating these latter linkages and analyzing the interactions in this network, we aim to showcase how woman-led collaborations are unique and how they impact firm-level invention performance. This two-network approach enables us to provide a nuanced analysis of gender roles in the technological sector and underscore diversity's importance in driving innovation outcomes.

This study utilizes firm-level panel data derived from patent records, which include the names of the inventors, the patent year, the number of citations, and other relevant details. This information allows us to aggregate and analyze data at the firm level in order to thoroughly understand each organization's innovation landscape. We employ a network analysis approach to examine firm collaboration patterns based on inventor affiliations. Each firm is represented as a node in the network, and an edge (a connection) is established between firms when their inventors jointly co-author a patent. This methodology enables us to visualize the intricate relationships that exist between firms in the innovation ecosystem. We explore the role of gender in collaborative invention by examining firms' collaborative ties based specifically on the gender of each patent's first author using our two gender-specific innovation networks mentioned above. If the first author of a co-authored patent is a man, the linkages between the authors' firms are included in the man-led innovation network. If the first author is a woman, the linkages are included in the woman-led innovation network. Distinguishing patenting teams by first author is fitting because existing studies (Koning et al., 2021) indicate women must occupy decision-making positions for gender to impact inventions. At the same time, when conducting regression analyses, we also run a robustness check by modeling co-invention networks based on the majority of inventors: if the majority of authors of a co-authored patent are men, the linkages between the authors' firms are included in the man-led innovation network and if the majority of authors are women, then linkages are included in the woman-led innovation network.

We analyze and compare the two gender-specific firm-level innovation networks to identify differences in collaboration patterns. We employed a five-year sliding window approach to delineate network characteristics in order to examine the evolving nature of firm-level

collaboration. This methodology, which is illustrated in Fig. 1, involves segmenting the overall network into sequential overlapping five-year subnetworks. The figure represents the progression of the sliding windows. Each subnetwork corresponds to a specific five-year period, with each subsequent subnetwork's window shifting forward one year. For instance, the subnetwork for the year 2004 encompasses all collaborations between firms that were documented from 2000 to 2004. Systematically shifting the window forward one year enables us to capture the longitudinal development of each network and observe the emergence, strengthening, and potential dissolution of connections over time. This fine-grained temporal analysis facilitates the calculation of precise metrics for each subnetwork to reveal changes in collaboration patterns and highlight key actors in specific time frames. Additionally, investigating how a firm's position in a network correlates with its co-patenting relationships in earlier subnetworks and its subsequent innovation outcomes enables us to assess how historical collaborative efforts influence future innovation success.

We systematically evaluate firms' network positioning based on their collaborative linkages. Each subnetwork's metrics are calculated using data about all collaborations that occurred over a five-year period. This comprehensive method ensures that our metrics reflect the cumulative effects of the links that firms have developed over time.

More formally, $N = \{1, 2, \dots, n\}$ is the set of n firms, L indicates the links between firms and therefore firms' co-patenting relationships, and $G(n \times n)$ is the network graph describing the adjacency matrix, where $a_{ij} = 1$ if $i \in N$ and $a_{ij} = 0$ otherwise. Each of the five-year subnetworks can be described as a graph $G_{interval} = (n, l, interval)$ in which $n \in N$ and $l \in L$ consider the firms and the links, respectively, in five-year intervals that change over time. Since our dataset covered the period from 1976 to 2019, forty subnetworks were created to compute the network metrics for each year from 1980 to 2019 based on the links that existed that year and in the preceding four years.

Analyzing these five-year subnetworks enabled us to effectively track and evaluate how a firm's past collaborative activities correlate with its subsequent innovation outcomes. The analysis was performed using panel data and made it possible to monitor changes and trends in innovation network dynamics over time.

3.1. Description of the variables

3.1.1. Dependent variables

3.1.1.1. Patent count. We use the number of patents granted to the inventors affiliated with each firm studied in a given year to measure patenting performance at the firm level.

3.1.1.2. Citation count. We also consider the number of citations that patents receive to measure patent quality. This enables us to analyze the impact, or quality, of innovation.¹ We examine the total number of three-year forward citations received by a firm's patents in a given year. This accounts for the time delay of citations, given that patents often take a few years to accrue citations and impact the research field. We also account for the fact it takes time for citations to accumulate by implementing a one-year lag in our analysis. This adjustment offers a

¹ To ensure consistency, we converted the unit of analysis of all the dependent variables from the individual level to the firm level.

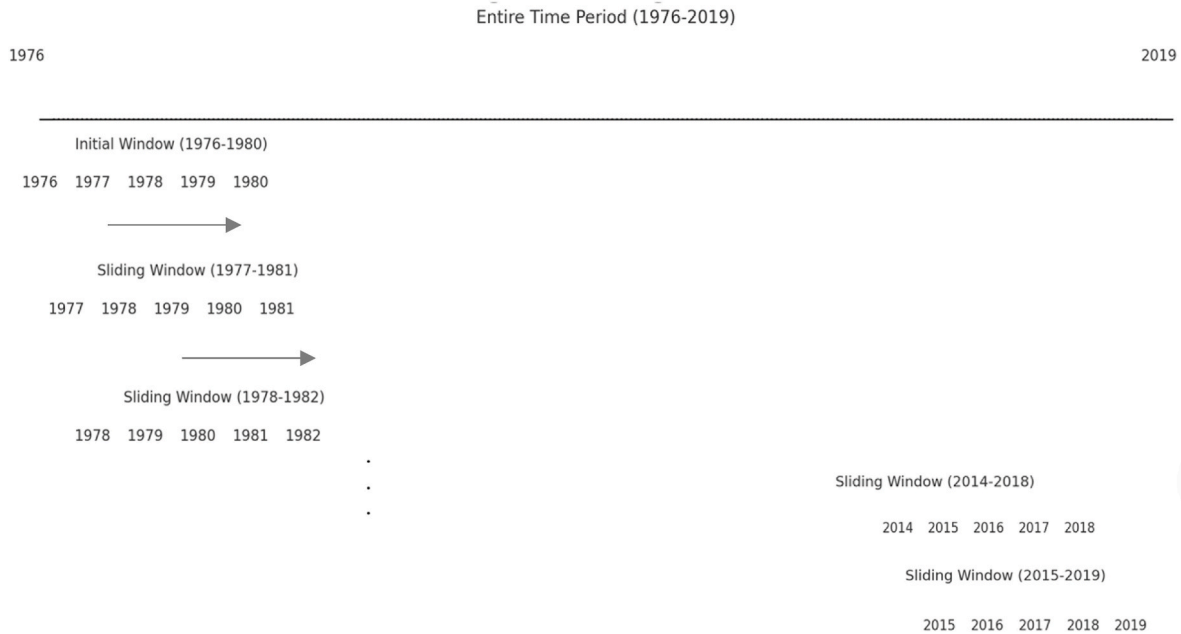


Fig. 1. – Visualization of the sliding window approach applied to subnetworks.

more precise representation of a patent's influence on subsequent innovation.

3.1.2. Independent variables

3.1.2.1. Boundary spanning -betweenness centrality. We use betweenness centrality² to measure boundary spanning, or brokerage. Incorporating betweenness centrality in our firm-level analysis of innovation networks offers a comprehensive understanding of firms' strategic positioning and collaboration dynamics. This centrality metric helps identify the key brokers in a network who facilitate interactions and the exchange of knowledge among firms. Moreover, firms with high betweenness centrality have considerable control over the flow of information and resources within a network, which highlights that central players play a pivotal role in shaping network dynamics. Measuring boundary spanning in a social network requires assessing the degree to which a node acts as a conduit between otherwise disconnected groups in a network and thereby controls the so-called structural holes in the network. Burt (1992) introduced various metrics to evaluate the presence and influence of structural holes. The betweenness index that was later developed by Burt et al. (2013) is used to quantify the extent to which a node has exclusive control over structural holes. Nodes that exhibit high betweenness centrality typically bridge structural holes and function as essential intermediaries that connect isolated segments of the network.

In our gender-specific innovation networks, firms that exhibit high betweenness centrality are structurally important because they play a pivotal role in shaping patenting productivity throughout the network. These firms are also strategically important because their presence ensures the continuity of connections with other firms. Consequently, betweenness centrality serves as a valuable indicator of firms' strategic

positioning in a network.

The betweenness centrality (B_c) of a given firm i is defined in Eq. (2), where $\sigma_{firm_j firm_k}$ represents the total number of shortest paths from firm j to firm k , and $\sigma_{firm_j firm_k}(firm_i)$ indicates the number of shortest paths from firm j to firm k that pass through firm i .

$$B_c(firm_i) = \sum_{i \neq j \neq k} \frac{\sigma_{firm_j firm_k}(firm_i)}{\sigma_{firm_j firm_k}} \quad \text{Eq. 2}$$

The betweenness centrality distributions for both gender-specific innovation networks are shown in Fig. 2.

The man-led innovation network has distinct nodes that have relatively higher betweenness centrality, which implies a more pronounced role in bridging different parts of the network. In contrast, the woman-led innovation network, despite also having key strategic nodes, does not display as high of centrality values, which suggests influence is more distributed among its nodes. The betweenness centrality distribution for the man-led innovation network reveals there is a concentration of firms with low centrality scores and the number of firms declines sharply as the centrality score increases. This pattern indicates that while many firms have limited influence in connecting different parts of the network, there are a few firms with much higher betweenness centrality. These firms act as key brokers and play crucial roles in facilitating communication and collaboration across the network. The betweenness centrality distribution for the woman-led innovation network also shows a high concentration of firms with low centrality scores, which indicates that most firms are not key brokers in the network. However, the firm count decreases steeply as centrality increases, and the centrality value range is narrower overall and has a lower maximum score in the woman-led innovation network than in the man-led one. This suggests that the woman-led innovation network has fewer firms that control the flow of information and collaboration and less-dominant firms than the man-led innovation network.

3.1.2.2. Clustering. Next, we determine each firm's clustering coefficient and triangle count in each of our gender-specific innovation

² In social network analysis, there are four standard node centrality metrics: degree centrality, which reflects the number of direct connections a node has and does not take into account indirect connections or the structure of the network; closeness centrality, which reflects the average shortest distance from each node to each other node; eigenvector centrality, which reflects connectedness to highly connected nodes; and betweenness centrality, which reflects node brokerage in the network. For the purpose of this study, we focus on betweenness centrality because it is the best indicator of boundary spanning.

³ Note that the two graphs have different y-axis scales because there is a different number of firms in each network.

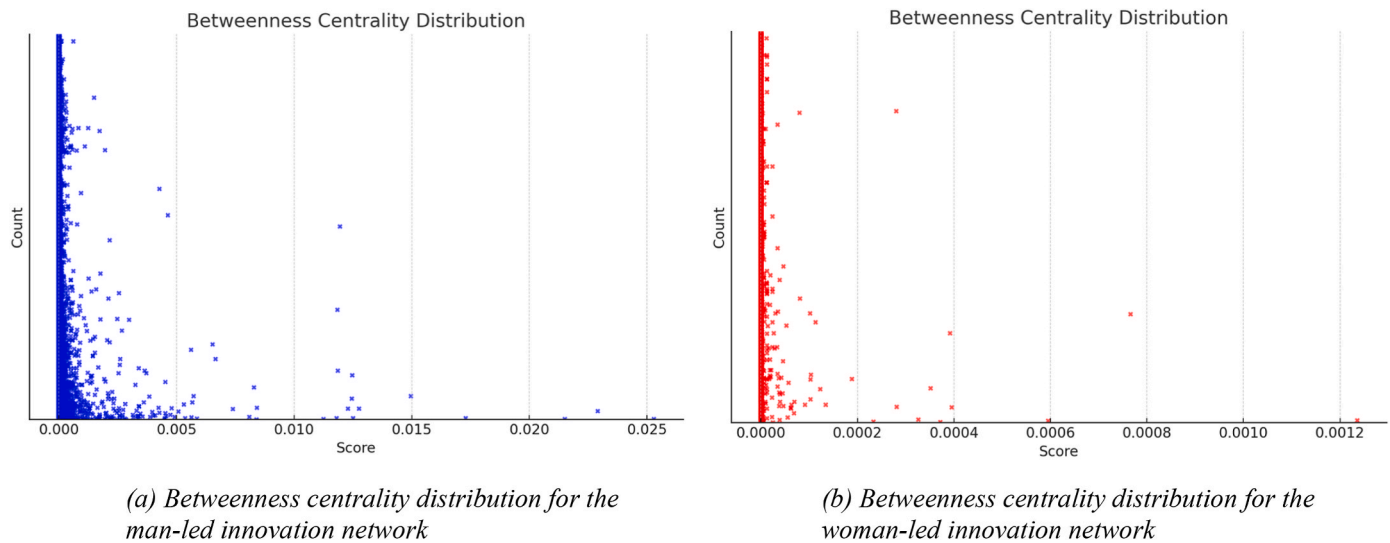


Fig. 2. – Betweenness centrality distributions of firms in the gender-specific networks.³

networks. Local cohesion is crucial for understanding how substructures develop and pinpointing which firms are influential in a network. The clustering coefficient sheds light on the interconnectedness of subgroups in a network by measuring the density of connections in a firm's immediate vicinity. It is expressed as the ratio of the number of connections that actually exist among a firm's neighbors to the total number of connections that are possible. A high clustering coefficient suggests a tight-knit group of firms and is indicative of strong local cohesion. We also consider an alternative clustering metric for robustness: the number of triangles—closed loops of reciprocated connections—that exist in a firm's neighborhood. Triangles are indicative of robust reciprocal relationships among a firm's neighbors and point to the existence of cohesive clusters in a network.

The clustering coefficient distributions for both gender-specific innovation networks are presented in Fig. 3.

In the man-led innovation network, there is a noticeable presence of firms with low clustering coefficients. This suggests that some firms work with collaborators who are less interconnected, which creates pockets of less cohesive collaboration. These firms with low clustering may serve as bridges between otherwise disconnected clusters, which indicates a more hierarchical network structure in which a few firms facilitate connections across the network. In contrast, in the woman-led innovation network, firms are distributed more uniformly across the entire range of clustering coefficient scores. This implies a higher degree of diversity in collaboration patterns, with firms demonstrating a wider range of collaboration structures. The more even distribution in the woman-led innovation network indicates that while many firms are part of tightly knit groups, there are also a significant number of firms with moderate levels of clustering. This reflects variation in collaboration dynamics and suggests a more egalitarian network structure in which influence and connectivity are more evenly distributed.

The diversity of clustering coefficients in the woman-led innovation network may contribute to increasing the resiliency and adaptability of the innovation ecosystem. Despite the fact that this network has fewer firms, its varied collaboration patterns can support dynamic and robust interactions, which foster a collaborative environment that can adapt to changes and challenges more effectively.

3.1.2.3. Network visualization by betweenness centrality and clustering.

⁴ Note that the two graphs have different y-axis scales because there is a different number of firms in each network.

We utilized Gephi, an open-source software designed for network visualization, to construct our networks and compute our network metrics. Gephi is equipped with an extensive suite of tools specifically geared towards dissecting the structural complexities of networks. This software makes it possible to thoroughly investigate the intricate dynamics and subtleties of network interactions.

Analyzing the giant component of the woman-led innovation network offers valuable insights into the structure of collaboration among woman-led patenting teams. This component corresponds to the largest connected subgraph and represents the core of the network where most interactions occur.

Betweenness centrality, which measures a firm's importance as a bridge between different parts of the network, reveals a pronounced hierarchy in the giant component (see Fig. 3a). Sony emerges as a key player with a higher betweenness centrality score than Fujitsu, which highlights its crucial role in facilitating collaboration and knowledge transfer between different firms. The clustering coefficient quantifies the degree to which a firm's neighbors are connected to each other, and its distribution reveals a network structure in which numerous firms exhibit high local connectivity (see Fig. 3b). These innovation hubs represent areas of intense collaboration that are likely characterized by knowledge and resource sharing. The fact that more firms have high clustering coefficients than high betweenness centrality scores suggests that the network structure supports local collaboration. Some firms are prominent in terms of their betweenness centrality but do not have high clustering coefficients. This indicates that these companies play crucial roles as brokers or bridges in the network and facilitate communication and the flow of resources between different clusters but are not heavily involved in tight-knit local collaboration hubs. Instead, their influence is more distributed because they act as pivotal points in the broader network rather than central figures in specific innovation clusters.

Fig. 4 showcases the five companies in our dataset that exhibit the highest levels of woman-led co-invention. The companies are: Prolifix, Fujisawa Pharmaceutical Co. Ltd, the Latvian Institute of Organic Synthesis, Teijin Limited, and AstraZeneca R&D. These firms stand out for their significant involvement of women inventors in decision-making roles, which underscores their commitment to fostering diversity within their respective industries (see Fig. 5).

3.1.3. Control variables

3.1.3.1. Average inventor share. Some innovation studies have argued that other factors not yet mentioned play a role in patenting

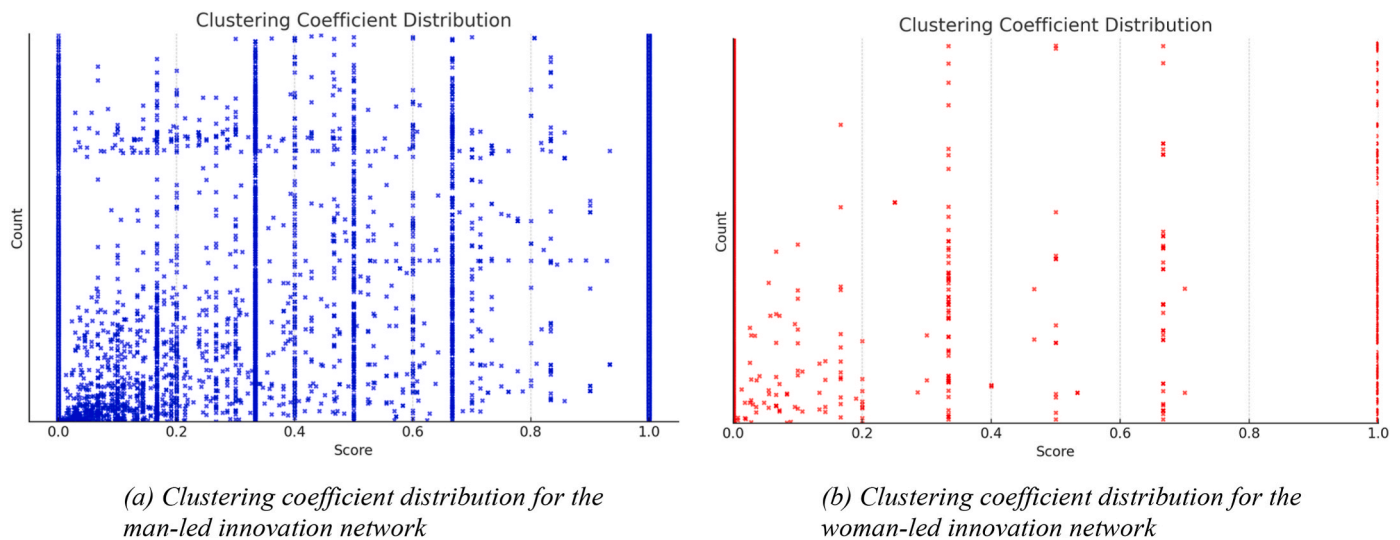


Fig. 3. Clustering coefficient distributions of firms in the gender-specific networks.⁴

performance. Previous studies (Beaudry and Kananian, 2013; Tahmooresnejad and Beaudry, 2019) have found average inventor productivity impacts patenting performance. The distribution of inventor share values varies from one co-invention relationship to the next, which reflects the collaborative dynamics at play between firms. To account for this variability, we include average inventor share as a control variable in our model. This variable is derived computationally by considering each inventor's contribution to each patent they co-author. More specifically, we calculate each inventor's share in each of their patents and then aggregate these shares to compute the average inventor share at the firm level. We therefore include in our model a variable that represents the average inventor share of all the inventors belonging to a given firm. The contribution of each inventor involved in each patent is indicated in the OECD REGPAT database used in this research.

3.1.3.2. Average inventor count. We also control for the average number of inventors per patent at the firm level. The extent of patent co-authorship is a salient indicator of collaboration in innovation activities. Empirical evidence suggests there is a positive correlation between the number of co-authors and patenting performance (Mirnezami et al., 2020; Tahmooresnejad and Turkina, 2022). Co-authorship fosters a

order to mitigate confounding factors and more accurately assess how network metrics are related to patenting performance.

3.1.3.5. Global cohesion. Our analysis also considers a variable that indicates how global cohesion impacts invention performance. This variable is calculated based on the network density of each subnetwork in our gender-specific innovation networks. A firm's global cohesion in each network refers to the degree to which the firm is connected to other firms and integrated in the network.

3.1.3.6. Country share. In addition, we include a country diversification metric to explore the potential benefits of international collaboration. This metric probes the breadth of firms' collaborative networks and dissemination of scientific knowledge across multiple countries. Our analysis of the international scope of firms' patenting activities seeks to provide insight into the patterns of knowledge exchange and innovation on an international scale.

3.1.3.7. Innovation efficiency. Finally, we control for a firm's ability to translate its inventor resources into patents over time using the firm-level variable *FirmInnovationEfficiency*, which is defined as:

$$\text{FirmInnovationEfficiency} = \frac{\text{Total Number of Patents}}{[(\text{Average Number of Inventors per Patent}) \times (\text{Number of Years of Active Patenting})]}$$

more rigorous development process and yields higher-quality patents.

3.1.3.3. Active patenting time. We consider the number of years that a firm has been actively patenting as a control variable to account for temporal effects and potential productivity issues. We utilize the year of a firm's first and last patents, which can be found in the OECD REGPAT database, to capture the temporal dimension and gain insight into the firm's patenting activity over time. This enables us to assess how the duration of a firm's patenting tenure influences its innovation outcomes.

3.1.3.4. Patent withdrawals and refusals. Furthermore, we consider the number of patents that a firm withdraws and has refused as proxies for low productivity, as they reflect instances when patent applications were either cancelled or rejected. We include these control variables in

This variable is designed to capture multiple dimensions of firm-level innovation:

- Firm R&D capacity

The numerator—Total Number of Patents—reflects a firm's overall innovation output. The denominator includes the firm's Average Number of Inventors per Patent, which serves as a proxy for its R&D team size.

- Time-normalized productivity

The denominator also incorporates the firm's Number of Years of Active Patenting to ensure that firms are compared fairly regardless of

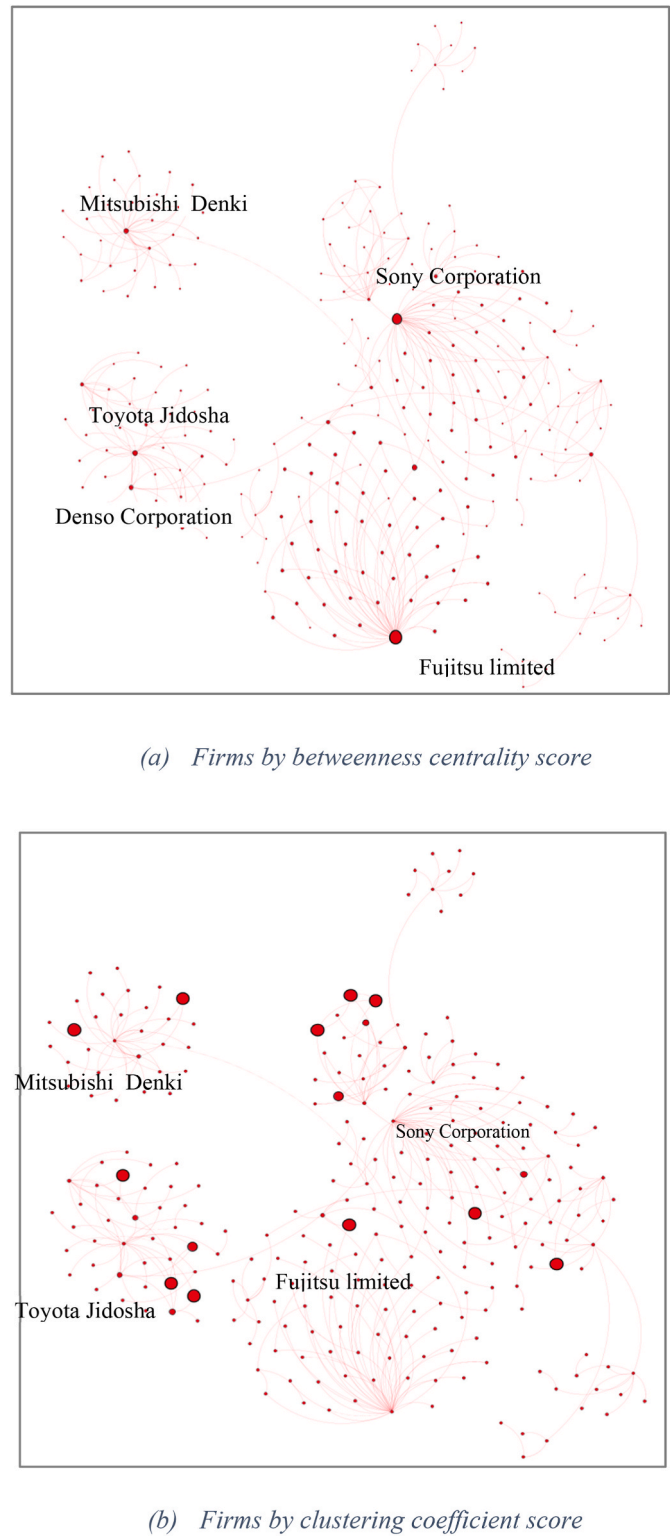


Fig. 4. – Firms in the giant component of the woman-led innovation network, by metric (higher scores are assigned bigger dots).

their operational history. This adjustment prevents longer-established firms from appearing more innovative merely because they have had more time to accumulate patents.

This variable measures how efficiently a firm converts its available inventor resources into patents over time. A higher value indicates that the firm in question produces a large number of patents relative to its inventor input and active patenting years, and signifies greater

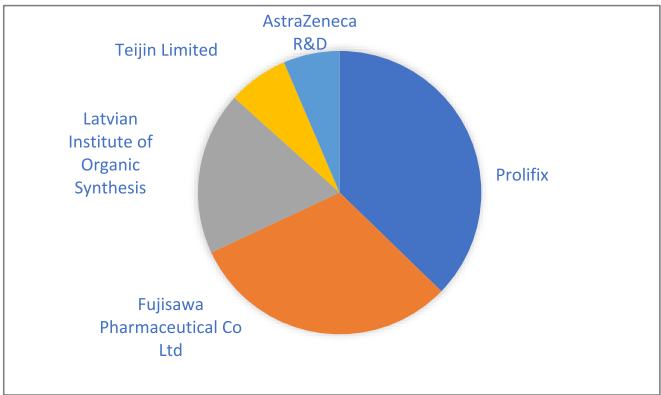


Fig. 5. – The five companies with the highest number of links in the woman-led innovation network.

Table 1
– Definition of the variables.

Variables	Description
Dependent variables	
<i>FirmPatt</i>	The total number of patents a firm receives in a given year
<i>FirmPatCitt</i>	The total number of 3-year forward citations that a firm's patents receive in a given year
Control variables	
<i>PatInventorShare_t</i>	The average inventor share of all the inventors affiliated with a firm in a given year
<i>PatActivityTime_t</i>	The number of years that a firm has been actively patenting, which denotes a firm's first patenting year and defines the duration by taking into account the number of years from the last patent averaged at the firm level
<i>PatInventorNum_t</i>	The average number of inventors per patent for each firm in a given year
<i>PatRefusal_t</i>	The proportion of the total number of patents refused divided by the total number of patents received by a firm in a given year
<i>PatWithdrawal_t</i>	The proportion of the total number of patents withdrawn divided by the total number of patents received by a firm in a given year
<i>PatCountryShare_t</i>	The average number of countries that a firm's inventors who patent are affiliated with in a given year
<i>FirmInnovationEfficiency_t</i>	The total number of patents produced by a firm divided by the product of the average number of inventors per patent and the number of years the firm has been actively patenting
<i>Dummy_year</i>	A dummy variable to account for the different years in a scholar's performance
Network variables	
<i>FirmBetweenness_t</i>	A firm's betweenness centrality in the man-led innovation network
<i>FirmClustering_t</i>	A firm's clustering coefficient in the man-led innovation network
<i>FirmTriangles_t</i>	A firm's triangle count in the man-led innovation network
<i>FirmGlobalCohesion_t</i>	A firm's global cohesion in the man-led innovation network
<i>FemFirmBetweenness_t</i>	A firm's betweenness centrality in the woman-led innovation network
<i>FemFirmClustering_t</i>	A firm's clustering coefficient in the woman-led innovation network
<i>FemFirmTriangles_t</i>	A firm's triangle count in the woman-led innovation network
<i>FemFirmGlobalCohesion_t</i>	A firm's global cohesion in the woman-led innovation network

innovation efficiency. Conversely, a lower value suggests inefficiency in converting inventor capacity into innovation outcomes. This variable comprehensively controls for firm-level heterogeneity.

All the variables used in this research are defined in Table 1.

3.2. Model

In our study, we employ negative binomial regression to analyze the count data. We chose to use this type of regression specifically to address the issue of over-dispersion that was evident in our preliminary data analysis because the variance significantly exceeded the mean. Negative binomial regression is particularly advantageous in these types of contexts as it incorporates an additional parameter to model the over-dispersion and is therefore more flexible when it comes to variance estimation than the more restrictive Poisson regression, which assumes equal mean and variance. This modeling approach uses a logarithmic link function that predicts the natural log of the expected count of the dependent variables from a linear combination of independent variables and includes a dispersion parameter to adjust for excessive variability. This application of negative binomial regression is well-suited to the empirical characteristics of our dataset, which exhibits clear signs of clustering and heterogeneity in its observations. This modeling approach enables us to effectively account for the count-based nature of our data while addressing the issue of over-dispersion in the dataset.

The regression model used in this study aims to analyze how firm-level patenting activity ($FirmPat_{it}$, $FirmPatCit_{it}$) is related to the network properties and control variables identified.

The model includes independent network variables for boundary spanning and clustering that have been transformed using natural logarithm functions to account for their skewed distributions and facilitate the interpretation of the results.⁵ Additionally, various control variables, one of which is a year dummy, are included to control for factors that may influence patenting activity. Our dataset's descriptive statistics are provided in [Appendices A1–A4](#) and the regression formulas are presented in [Appendix A5](#).

3.2.1. Model estimation

[Tables 2–5](#) detail the results of our panel negative binomial regressions analyzing how network metrics influence firms' innovation outcomes, such as their patent and citation counts, in both gender-specific innovation networks. [Tables 2 and 3](#) focus on the man-led innovation network and indicate how metrics like betweenness centrality and local cohesion impact firms' invention performance in that network. [Tables 4 and 5](#) present the corresponding analysis results for the woman-led innovation network to provide insight into how the metrics affect firms' invention performance in that network. We selected three models for each dependent variable considered and confirmed that multicollinearity is not an issue in the models.⁶

3.2.2. Model estimation for the man-led invention network

When it comes to the impact network metrics have on a firm's patent output in the man-led innovation network, our results in [Table 2](#) reveal that betweenness centrality is positively correlated with patent activity, which indicates that firms that act as brokers in the man-led innovation network and connect its isolated groups drive innovation by facilitating the exchange of knowledge and resources. This suggests that firms with higher betweenness centrality play a crucial role in this network by serving as bridges between otherwise disconnected groups. These firms are likely to have access to a wide range of information and ideas, which they can combine and leverage to generate novel solutions and technologies. Furthermore, their ability to connect different parts of the network enables them to mobilize resources more effectively and create

synergies that can lead to increased patenting activity.

The regression results indicate how the clustering coefficient and triangle count, which are local network cohesion metrics, influence the number of patents a firm produces. Our analysis results show that these network metrics negatively impact innovation outcomes in the man-led innovation network. In this network, having a higher clustering coefficient, which is indicative of a firm having a greater proportion of closed triadic relationships in its network, is associated with having fewer patents. This suggests that excessive local cohesion leads to information redundancy and having limited exposure to novel ideas, which hinder innovation in the man-led innovation network. Similarly, having a higher triangle count is associated with reduced patent productivity and patent quality (as measured by the number of patent citations) in this network.

The control variables used in our regressions offer valuable insights into factors that influence patent production in the man-led innovation network. Notably, *PatActivityTime* is consistently very positively associated with patent output across all the models considered. This suggests that the amount of time invested in patent-related activities, such as R&D, patent filing, and intellectual property management, is a robust predictor of patent productivity in this network. Firms that allocate more resources and time to these processes may cultivate a more robust innovation pipeline and more efficiently navigate the complexities of the patenting process, and ultimately improve their innovation outcomes. In this network, the share of each patent that is attributed to its individual inventors (*PatInventorShare*) positively impacts invention performance. When co-inventors have a higher inventor share in their patents they appear more likely to produce additional patents in the future. This suggests that greater individual contributions to collaborative efforts can lead to increased productivity and sustained innovation over time for firms in this network.

Conversely, *PatWithdrawal* and *PatRefusal* are consistently very negatively associated with patent output across all the models considered. High rates of patent withdrawal and refusal may be indicative of inefficiencies or challenges in the patent application process. Patent withdrawals can be due to strategic decisions, perceived lack of patent quality, or financial limitations, while refusals often signal that patents do not meet patentability criteria or lack novelty. These negative outcomes highlight potential areas for organizational improvement, including enhancing the quality of patent applications, refining strategic patent portfolio management, and ensuring sufficient resources are allocated to the patenting process. Furthermore, the fact that *PatInventorNum* is positively associated with patent output underscores how important collaborative efforts and diverse expertise are in driving innovation. Having a greater number of inventors involved in the patenting process may reflect a broader range of skills, perspectives, and knowledge bases, and lead to the development of more innovative and higher-quality patents. Firms that foster a collaborative environment and encourage contributions from multiple inventors are likely to have access to a wider range of ideas and approaches and ultimately enhance their innovation capacity and patent output. In addition, *FirmGlobalCohesion* is negatively associated with patent productivity in some models. This suggests that a firm's global cohesion can hinder its innovation outcomes.

[Table 3](#) presents the regression results that indicate how firms' network positioning in the man-led innovation network is related to their innovation quality as measured by the citation impact of their patents over a three-year period. The betweenness centrality results indicate that a firm's strategic positioning in the man-led innovation network significantly influences the citation performance of its patents. Firms with higher betweenness centrality in this network leverage their brokerage role to enhance the visibility and impact of their patents. Acting as intermediaries enables these firms to access new information and innovations from different parts of the network and disseminate them to other parts. This brokerage role facilitates the flow of knowledge and enhances the firm's reputation within the community, which

⁵ We also considered a one-year lag for the network metrics to account for the fact it takes time for collaborations to impact performance. However, our results are more robust without this lag, as the delayed effects that network dynamics have on innovation outcomes are already captured by using five-year sliding windows when constructing the subnetworks.

⁶ We also conducted a preliminary correlation analysis, which revealed that the independent variables in each of the models are only weakly correlated.

Table 2– The regression results of the impact of Betweenness centrality and Clustering on the number of patents in man-led network.^a.

Variables	Models					
<i>FirmPat_t</i>	Model 1		Model 2		Model 3	
Network variables						
<i>FirmBetweenness_t</i>	0.034 (0.001)	***				
<i>FirmTriangles_t</i>			−0.008 (0.001)	***		
<i>FirmClustering_t</i>					−0.104 (0.011)	***
Control variables						
<i>PatInventorShare_t</i>	0.367 (0.015)	***	0.372 (0.014)	***	0.381 (0.014)	***
<i>PatActivityTime_t</i>	0.452 (0.013)	***	0.547 (0.011)	***	0.553 (0.011)	***
<i>PatWithdrawal_t</i>	−0.517 (0.020)	***	−0.522 (0.017)	***	−0.528 (0.017)	***
<i>PatRefusal_t</i>	−0.150 (0.020)	***	−0.147 (0.017)	***	−0.149 (0.017)	***
<i>PatCountryShare_t</i>	−0.048 (0.017)	***	−0.061 (0.012)	***	−0.059 (0.012)	***
<i>PatInventorNum_t</i>	0.804 (0.016)	***	0.786 (0.014)	***	0.793 (0.014)	***
<i>FirmGlobalCohesion_t</i>	−0.937 (0.092)	***	−1.340 (0.100)	***		
<i>FirmInnovationEfficiency</i>	0.547 (0.008)	***	0.627 (0.008)	***	0.636 (0.008)	***
<i>FirmClustering_t</i> [']					−0.002 (0.003)	
<i>FirmGlobalCohesion_t</i>						
<i>Dummy_year</i>	Yes		Yes		Yes	
<i>Constant</i>	9.594 (0.912)	***	15.571 (1.162)	***	0.104 (0.124)	
<i>lnalpha_constant</i>	−0.635 (0.015)	***	−0.606 (0.013)	***	−0.597 (0.013)	***
<i>Nb of observations</i>	37298		37298		37298	
<i>Nb of groups</i>	25335		25335		25335	
<i>R² pseudo</i>	0.2386		0.2307		0.2296	
<i>Loglikelihood</i>	−37972		−47439		−47508	

Note: ***, ** and * show significance at the 1 %, 5 % and 10 % levels, respectively, and standard errors are presented in parentheses.

^a Transformed variables are used in the regression models to improve model fit.

increases the likelihood of its patents being cited.

Further analysis reveals that *PatActivityTime* has a robust positive association with patent citation count in this network across all the models considered. This finding suggests that investing more time and resources in patent-related activities is positively correlated with increased recognition of patent value in the scientific and technological community. Firms that engage more extensively in these endeavors are more likely to produce patents that garner greater scholarly attention and have more impact. Conversely, *PatWithdrawal* and *PatRefusal* are very negatively associated with patent citation count in this network. Elevated withdrawal and refusal rates may indicate underlying deficiencies in the patent application process or potential shortcomings in terms of the quality of submitted patents. Withdrawals and refusals are both correlated with reduced scholarly interest and impact.

As for our *PatInventorShare* variable, when individual inventors are attributed a greater share of the contributions made to their patents, it negatively impacts their citation counts. This may indicate that patents heavily attributed to a single inventor or a small group may not receive as much recognition and as many citations as those with more balanced contributions from multiple inventors. Authorship diversity likely brings with it different perspectives and expertise, which can enhance the quality and relevance of a patent and lead to higher citation counts.

Our *PatCountryShare* variable is positively related to patent citation count in this network despite being associated with a slight decrease in patent count. This observation suggests that having a higher proportion of patents involving inventors from different countries or regions may

not necessarily impede, and could actually increase, a firm's overall scholarly impact and recognition of its intellectual property. This phenomenon implies that patents that are developed by inventors from different countries may be perceived as more valuable or innovative, perhaps due to factors such as the robustness of the local innovation ecosystem, the reputation of research institutions in the region, or the specific technological focus that is prevalent in the area. Firms may therefore strategically benefit from involving inventors from various countries to enhance the visibility and citation impact of their patented innovations.

PatInventorNum, for its part, has a similar impact on patent citation count as patent productivity: it is very positively associated with a firm's citation impact, which underscores how pivotal collaboration and diverse expertise are when it comes to enhancing patent quality and visibility. Having a larger number of inventors contributing to the patenting process may signify the team has a broader range of skills, perspectives, and knowledge bases, and potentially lead to the development of more comprehensive and impactful inventions. Patents resulting from such collaborative efforts may be perceived as more robust and relevant in the scientific community and therefore garner more citations. Interestingly, *FirmGlobalCohesion* has a negative effect on the number of patents and is insignificant with citations, overall, supporting the evidence that clustering is not beneficial for firms in the man-led inventor network.

FirmInnovationEfficiency is positive and significant in all the models, indicating a direct positive impact on patent productivity and impact.

Table 3

– The regression results of the impact of Betweenness centrality and Clustering on the number of patent citations in man-led network.

Variables	Models					
<i>FirmPatCitt</i>	Model 1		Model 2		Model 3	
Network variables						
<i>FirmBetweenness_t</i>	0.118 (0.015)	***				
<i>FirmTriangles_t</i>			–0.014 (0.003)	***		
<i>FirmClustering_t</i>					–0.142 (0.058)	**
Control variables						
<i>PatInventorShare_t</i>	–0.100 (0.030)	***	–0.097 (0.030)	***	–0.083 (0.037)	**
<i>PatActivityTime_t</i>	0.456 (0.031)	***	0.522 (0.030)	***	0.551 (0.032)	***
<i>PatWithdrawal_t</i>	–0.369 (0.036)	***	–0.365 (0.036)	***	–0.375 (0.042)	***
<i>PatRefusal_t</i>	–0.114 (0.037)	***	–0.112 (0.037)	***	–0.122 (0.051)	**
<i>PatCountryShare_t</i>	0.105 (0.030)	***	0.108 (0.030)	***	0.124 (0.027)	***
<i>PatInventorNum_t</i>	0.517 (0.030)	***	0.535 (0.029)	***	0.556 (0.037)	***
<i>FirmGlobalCohesion_t</i>	0.184 (0.224)					
<i>FirmInnovationEfficiency</i>	0.636 (0.022)	***	0.678 (0.022)	***	0.704 (0.023)	***
<i>FirmClustering_t × FirmGlobalCohesion_t</i>					–0.237 (0.091)	***
<i>Dummy_year</i>	Yes		Yes		Yes	
<i>Constant</i>	–2.716 (2.458)		–1.016 (0.125)	***	0.191 (0.098)	*
<i>lnalpha_constant</i>	1.459 (0.016)	***	1.462 (0.016)	***	1.435 (0.016)	***
<i>Nb of observations</i>	37298		37298		37298	
<i>Nb of groups</i>	25335		25335		25335	
<i>R² pseudo</i>	0.0884		0.088		0.0862	
<i>Loglikelihood</i>	–33903		–33918		–32571	

Note: ***, ** and * show significance at the 1 %, 5 % and 10 % levels, respectively, and standard errors are presented in parentheses.

Our analysis of how network metrics impact patenting performance in the man-led innovation network highlights how critical strategic networking and positioning are to collaborative networks in terms of both patent and citation count. Firms that occupy boundary-spanning positions in the man-led innovation network are more likely to see their patents receive more citations. However, excessive cohesion hinders innovation in this network. These results support our first and third hypotheses.

3.2.3. Model estimation for the woman-led invention network

This section of our paper investigates how having women lead patenting teams shapes firm-level innovation outcomes. The results of our regression analyses for the woman-led innovation network are detailed in Tables 4 and 5. Contrary to our Hypothesis 2, betweenness centrality is significant in predicting patents and patent citations in the woman-led network. Therefore, in the woman-led innovation network, firms that are strategically positioned as brokers (have high betweenness centrality) exhibit greater patent productivity and patent impact. This finding highlights the importance of strategic networking in facilitating the exchange of knowledge and access to resources, which turn out to be important for successful invention in the woman-led network too. At the same time, the test of difference of coefficients (Appendix A6) for betweenness centrality indicates that the coefficients differ in a statistically significant manner between the woman-led and man-led innovation networks, which means that while betweenness centrality is significant in woman-led network, it has a stronger impact in the man-led innovation network both in the case of the number of patents and

in the case of patent citations.

The results presented in Table 4 show that while clustering has a weak influence on the number of patents, the other clustering metric considered—triangle count—has a stronger positive effect. We also explored the effects of the interaction variable of clustering and global cohesion, which turned out positive and significant indicating that local clustering is important when there is global cohesion: strong local connections and global integration can synergistically enhance firms' innovation outcomes in the woman-led innovation network. Overall, these findings emphasize that close-knit subgroups and cohesive local and global connections play a critical role in fostering a fertile environment for inventions in the woman-led innovation network. Cohesion facilitates knowledge sharing, collaboration, and mutual support, and ultimately spurs the creation of patentable inventions.

As for the control variables, effective patent management practices, particularly ones that aim to minimize patent withdrawals and refusals, are instrumental in optimizing patent output. Firms can maximize their innovation potential by ensuring patent applications are strategically aligned and of high quality. The negative regression results observed for *PatWithdrawal* and *PatRefusal* in this network reinforce the importance of reducing patent withdrawals and refusals to enhance a firm's patent productivity.

Additionally, fostering a collaborative environment that values different perspectives and encourages contributions from multiple inventors is likely to enhance patent productivity in this network, as is evidenced by the positive results associated with *PatInventorNum*. This finding highlights that collaborative effort and invention output are

Table 4

– Regression results indicating the impact of betweenness centrality and clustering on the number of patents in the woman-led innovation network.

Variables	Models		Model 2		Model 3	
<i>FirmPat_t</i>	Model 1					
Network variables						
<i>FemFirmBetweenness_t</i>	0.026 (0.004)	***				
<i>FemFirmTriangles_t</i>			0.063 (0.030)	**		
<i>FemFirmClustering_t</i>					0.168 (0.099)	*
Control variables						
<i>PatInventorShare_t</i>	0.354 (0.038)	***	0.363 (0.039)	***	0.372 (0.040)	***
<i>PatActivityTime_t</i>	0.349 (0.030)	***	0.385 (0.029)	***	0.408 (0.029)	***
<i>PatWithdrawal_t</i>	−0.285 (0.035)	***	−0.287 (0.036)	***	−0.300 (0.036)	***
<i>PatRefusal_t</i>	−0.085 (0.033)	***	−0.093 (0.033)	***	−0.092 (0.033)	***
<i>PatCountryShare_t</i>	0.005 (0.027)		0.028 (0.028)		0.015 (0.029)	
<i>PatInventorNum_t</i>	0.560 (0.039)	***	0.594 (0.040)	***	0.598 (0.041)	***
<i>FemFirmGlobalCohesion_t</i>	0.017 (0.040)		0.035 (0.004)	***		
<i>FirmInnovationEfficiency</i>	0.528 (0.014)	***	0.557 (0.014)	***	0.545 (0.014)	***
<i>FirmClustering_t</i>					0.086 [*] (0.064)	
<i>FemFirmGlobalCohesion_t</i>						
<i>Dummy_year</i>	Yes		Yes		Yes	
Constant	−0.550 (0.261)	**	0.679 (0.031)	***	0.186 (0.129)	
<i>lnalpha_constant</i>	−1.295 (0.053)	***	−1.181 (0.050)	***	−1.247 (0.053)	***
<i>Nb of observations</i>	5375		5375		5375	
<i>Nb of groups</i>	4249		4249		4249	
<i>R²_pseudo</i>	0.2887		0.2787		0.2881	
<i>Loglikelihood</i>	−4242		−4302		−4068	

Note: ***, ** and * show significance at the 1 %, 5 % and 10 % levels, respectively, and standard errors are presented in parentheses.

positively correlated, which suggests that involving a greater number of inventors in woman-led patenting teams can lead to more patents. Interestingly, the significant positive effect that global cohesion has specifically in the woman-led innovation network suggests that promoting global collaboration may be particularly advantageous for firms in this network. This implies that fostering a globally cohesive environment can lead to greater innovation and productivity among woman-led patenting teams and potentially bridge gender gaps in patenting and innovation.

As for the other innovation metric, citation count, both clustering metrics (clustering coefficient and triangle count) have a positive impact on patent quality and visibility. Overall, these results support our fourth hypothesis regarding clustering being positively associated with invention performance in the woman-led innovation network. These findings show that taking a gender-based approach that involves comparing the firm-level results obtained in the woman-led and man-led innovation networks is justified.

In addition, we observe that geographical diversity contributes significantly to receiving more citations in the woman-led innovation network. The positive results for *PatCountryShare* indicate that patents that have a woman as the first author and a geographically diverse team are more likely to be cited. This suggests that international diversification enhances the visibility and recognition of patents in the woman-led innovation network. Strategic patent management, including minimizing the number of patent withdrawals and refusals, is also essential for enhancing a firm's citation performance in this network. Moreover,

Table 5

– Regression results indicating the impact of betweenness centrality and clustering on the number of patent citations in the woman-led innovation network.

Variables	Models		Model 2		Model 3	
<i>FirmPat_tCit</i>	Model 1					
Network variables						
<i>FemFirmBetweenness_t</i>	0.030 (0.009)	***				
<i>FemFirmTriangles_t</i>			0.100 (0.060)	*		
<i>FemFirmClustering_t</i>					0.600 (0.402)	*
Control variables						
<i>PatInventorShare_t</i>	−0.103 (0.072)		−0.114 (0.074)		−0.094 (0.074)	
<i>PatActivityTime_t</i>	0.104 (0.074)		0.016 (0.073)		0.189 (0.073)	***
<i>PatWithdrawal_t</i>	−0.324 (0.075)	***	−0.274 (0.080)	***	−0.334 (0.077)	***
<i>PatRefusal_t</i>	0.020 (0.059)		0.023 (0.061)		0.021 (0.060)	
<i>PatCountryShare_t</i>	0.154 (0.063)	**	0.217 (0.070)	***	0.156 (0.067)	**
<i>PatInventorNum_t</i>	0.365 (0.071)	***	0.411 (0.074)	***	0.397 (0.073)	***
<i>FemFirmGlobalCohesion_t</i>	0.092 (0.098)					
<i>FirmInnovationEfficiency</i>	0.633 (0.050)	***	0.720 (0.061)	***	0.656 (0.052)	***
<i>FirmClustering_t</i>					0.201 (0.239)	
<i>FemFirmGlobalCohesion_t</i>						
<i>Dummy_year</i>	Yes		Yes		Yes	
Constant	−2.527 (0.756)	***	0.724 (0.073)	***	−1.904 (0.534)	***
<i>lnalpha_constant</i>	0.990 (0.046)	***	1.138 (0.045)	***	1.019 (0.047)	***
<i>Nb of observations</i>	5375		5375		5375	
<i>Nb of groups</i>	4249		4249		4249	
<i>R²_pseudo</i>	0.0849		0.0642		0.0831	
<i>Loglikelihood</i>	−3926		−4015		−3717	

Note: ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively, and standard errors are presented in parentheses.

involving more inventors significantly boosts citation count, which underscores how powerful collaborative innovation and varied expertise are when it comes to producing influential, highly-cited patents.

The duration of a firm's active patenting period positively influences its invention performance and the citation impact of its patents in both gender-specific innovation networks. In addition, firms that have a high number of withdrawn or refused patents exhibit lower patenting performance in both networks, which confirms it is important to ensure patent applications are of high quality and strategically aligned.

We also conduct a robustness check and redo all the regressions modeling the networks based on the share of men and women in patent inventor teams instead of lead author being a man or a woman. The results are presented in Appendix A7. The results support our main findings.

4. Discussion and conclusion

4.1. Theoretical and empirical contributions

Our research contributes to the literature on innovation networks by studying the impact of two key network properties—boundary spanning and clustering—on firms' invention performance and by introducing gender as an important boundary condition for this link. In doing so, we contribute to the growing body of literature (Hall and Bagchi-Sen, 2007; Crescenzi et al., 2015; Lv et al., 2020) that examines factors that drive firms' invention performance. We bridge the gap between studies on

inventor networks (Toth and Lengyel, 2021; Le Gallo and Plunket, 2020), studies that explore how gender impacts innovation (Galasso and Simcoe, 2011; Nielsen and Nielsen, 2013; Garcia Martinez et al., 2017; Arun and Rojers, 2021), and studies that focus on gender and social networks (Brands et al., 2022). By specifically examining the interplay between gender, networking, and firms' invention performance, this paper directly addresses the research shortcomings highlighted by Agnete Alsos et al. (2013) and Arun and Rogers (2021). Our analysis delves into firms' interactions on the basis of their inventors' co-patenting relationships and explains the advantages network patterns provide with a focus on two critical indicators of invention performance: innovation productivity, as measured by firms' patent count, and innovation quality, as measured by the citation impact of firms' patents.

Being able to understand and leverage the dynamics at play in inter-firm innovation networks will enable firms to position themselves more strategically, enhance their innovation capability, and maintain a competitive edge. While previous studies have established that innovation networks play a significant role in firms' invention performance (Le Gallo and Plunket, 2020), the literature is inconsistent with regard to how two key network positions—boundary spanning and clustering—affect firms' innovation. Some studies have argued that boundary spanning is beneficial for firms' invention performance and clustering impedes it, while other studies have found clustering has a positive effect on it. Our study helps to nuance that it is important to consider the role of gender in co-invention and that how network positions affect firms' invention performance is gender-dependent. In man-led inter-firm innovation networks, firms benefit most from being in a boundary-spanning position, as our betweenness centrality results reveal, while clustering negatively affects firms' invention performance. Boundary spanning boosts firms' innovation productivity in these networks by giving them access to a variety of knowledge and resources that facilitate innovation in the long run. Clustering, on the other hand, hinders firm-level innovation in these networks by creating a lock-in effect.

In woman-led inter-firm innovation networks, firms benefit from being in a boundary-spanning position, but to a lesser extent than in man-led innovation networks, and are also positively impacted by clustering, which is not the case in man-led innovation networks. Collaborating in tight-knit, highly interconnected groups, as well as in globally cohesive networks, enhances the performance and visibility of woman-led innovations and helps improve their citation impact. This result aligns with studies that suggest that women inventors tend to repeat their past collaborations (Ductor et al., 2023; Lindenlaub and Prummer, 2021), feel more comfortable in groups with high levels of trust and interconnectedness (Szell and Thurner, 2013), and are more productive in environments where social support and trust are prevalent, which is often the case in repeat interactions. Conversely, men inventors tend to focus more on general innovations (Nählinger, 2010; Blake and Hanson, 2005), prioritize gaining efficiency and results from their connections, and benefit more from having long-reach connections in their networks. These findings are in line with recent studies on friendship networks (Brands and Mehra, 2019) that found there were gender-based differences in broker performance in networks and revealed that (a) women and men psychologically construe brokerage in networks differently and (b) this difference helps to account for gendered differences in broker performance. Women (but not men) perform worse when they perceive themselves as brokers in networks (Brands and Mehra, 2019). We observe a similar trend in the gender-specific innovation networks that we studied: brokerage, as measured by betweenness centrality, has a weaker positive effect in our woman-led innovation network than in our man-led innovation network. Additionally, brokerage has a weaker effect on citation count than clustering does.

Our results have important practical implications. Firms that wish to increase their invention performance should pay attention to their network position in inventor networks and to the gender aspects of their

inventor teams. In man-led innovation networks, it is beneficial to encourage boundary-spanning connections; prioritize connecting with new collaborators, international teams, and inventors in other domains; and avoid clustering, which will create a lock-in situation and foster isomorphism. In woman-led innovation networks, on the other hand, it is beneficial to encourage repeat interactions and collaboration with firms that have previously worked together and co-patented. This builds trust and social capital, which enables woman-led patenting teams to be more innovative. It is important to note, however, that boundary-spanning connections do positively impact firms' invention performance in woman-led innovation networks, though they do so to a lesser extent than in the man-led innovation network. This indicates that it may be important for woman-led innovation networks to have some long-reach connections in order to access new knowledge, ideas, resources. These findings underscore the importance of balancing local cohesion with broader network connections to foster innovation. As previously mentioned, women tend to benefit less from brokering than men do because they perceive brokering positions to be not worth it or too ambitious or bold for them. Firms could work with their women inventors' perceptions, explain to them the value of brokering, and make an extra effort to encourage them to participate in more distant collaborations. With time, this may help women inventors to change their perception of brokerage and extract more value from brokering positions. Our study demonstrates that women inventors in decision-making positions (as well as overall in inventor teams as demonstrated by our robustness check) play a crucial role in inter-firm networking and improve firms' invention performance because their networking strategies positively affect invention performance. Therefore, it is beneficial for firms to encourage their women inventors to lead patenting teams and to generally participate in patenting teams, and firms can use our findings to help their women inventors become more prolific with regard to patent productivity and patent quality.

4.2. Limitations and future research directions

Future research could provide more insights into gender, network structures and innovation by focusing on differences that are specific to technological disciplines, as this could further illuminate the specific network dynamics that are at play in woman-led and man-led innovation networks and how they affect firms' invention performance. It is possible that women scientists in certain disciplines have a greater proclivity to assume lead author positions and show heightened interest in these positions.

As a final note, this study has its limitations. First, our results are limited to the sample of firms and data that we extracted from the OECD REGPAT database. We used firms' patenting activity to measure the collaborative links that exist among their affiliated inventors; however, firms do develop innovations that they do not patent or disclose. Future research is expected to analyze the other innovation relationships that exist between firms to verify whether similar findings are obtained for the network variables. Additionally, future studies could distinguish between different types of inter-firm collaboration (alliances, acquisitions, joint ventures, etc.) and explore whether the type of collaboration has implications for the effects of network properties in woman-led and man-led innovation networks.

Moreover, while the OECD REGPAT database's comprehensive inventor and patent data did benefit our study, it is important to note one limitation the database has: it lacks firm-level information about inventors' average patenting tenure. This omission is noteworthy, as an inventor's prior patenting experience can significantly impact a firm's invention performance.

CRedit authorship contribution statement

Leila Tahmooresnejad: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis,

Data curation, Conceptualization. **Ekaterina Turkina**: Writing – review & editing, Writing – original draft, Visualization, Validation,

Supervision, Methodology, Investigation, Funding acquisition, Conceptualization.

Appendix A1. Descriptive statistics for the firm-level man-led invention network

Variables	Obs	Mean	Std. Dev.	Min	Max
<i>FirmPat_t</i>	37,298	3.338	5.831	1	186
<i>FirmPatCit_t</i>	37,298	1.695	7.077	0	352
<i>FirmDegCentrality_t</i>	37,298	1.416	1.865	0	32
<i>FirmBetweenness_t</i>	37,298	0.000	0.000	0	0.008
<i>FirmClustering_t</i>	37,298	0.268	0.423	0	1
<i>FirmTriangles_t</i>	37,298	0.794	3.281	0	147
<i>PatInventorNum_t</i>	37,298	3.099	1.630	1	38
<i>PatActivityTime_t</i>	37,298	0.341	0.839	0	20
<i>PatWithdrawal_t</i>	37,298	0.205	0.325	0	1
<i>PatRefusal_t</i>	37,298	0.018	0.107	0	1
<i>PatCountryShare_t</i>	37,298	1.028	0.172	1	3
<i>FirmGlobalCohesion_t</i>	37,298	0.000	0.001	0	0.044
<i>PatInventorShare_t</i>	37,298	1.333	0.318	0.184	2.07
<i>FirmInnovationEfficiency</i>	37,298	3.240	17.619	0.01	1312.3

Appendix A2. Descriptive statistics for the firm-level woman-led invention network

Variables	Obs	Mean	Std. Dev.	Min	Max
<i>FirmPat_t</i>	5375	3.559	20.080	1	720
<i>FirmPatCit_t</i>	5375	2.391	9.492	0	256
<i>FemFirmDegCentrality_t</i>	5375	1.101	2.063	0	28
<i>FemFirmBetweenness_t</i>	5375	0.000	0.000	0	0.001
<i>FemFirmClustering_t</i>	5375	0.034	0.164	0	1
<i>FemFirmTriangles_t</i>	5375	0.194	1.085	0	21
<i>PatInventorNum_t</i>	5375	4.262	2.407	1	50
<i>PatActivityTime_t</i>	5375	0.292	0.907	0	20
<i>PatWithdrawal_t</i>	5375	0.301	0.399	0	1
<i>PatRefusal_t</i>	5375	0.025	0.136	0	1
<i>PatCountryShare_t</i>	5375	1.064	0.279	1	5
<i>FirmGlobalCohesion_t</i>	5375	0.001	0.001	0	0.034
<i>PatInventorShare_t</i>	5375	1.1620	0.2932	0.685	2.07
<i>FirmInnovationEfficiency</i>	5375	2.3317	8.729	0.006	136.6

Appendix A3. Correlation matrix for the firm-level man-led invention network

	1	2	3	4	5	6	7	8	9	10	11	12		
<i>FirmPat_t</i>	1	1												
<i>FirmPatCit_t</i>	2	0.426	1											
<i>FirmBetweenness_t</i>	3	0.229	0.082	1										
<i>FirmClustering_t</i>	4	−0.076	−0.012	0.081	1									
<i>FirmTriangles_t</i>	5	0.003	0.029	0.323	0.784	1								
<i>PatInventorNum_t</i>	6	0.134	0.087	0.044	0.315	0.343	1							
<i>PatActivityTime_t</i>	7	0.309	0.106	0.333	−0.061	0.011	−0.131	1						
<i>PatWithdrawal_t</i>	8	−0.177	−0.067	−0.113	0.083	0.071	−0.016	−0.263	1					
<i>PatRefusal_t</i>	9	−0.045	−0.010	−0.024	0.025	0.022	−0.001	−0.061	−0.082	1				
<i>PatCountryShare_t</i>	10	−0.045	0.004	−0.026	0.068	0.071	0.038	−0.083	0.056	0.019	1			
<i>FirmGlobalCohesion_t</i>	11	−0.003	−0.034	0.020	−0.037	−0.017	−0.008	0.056	−0.057	0.008	−0.028	1		
<i>PatInventorShare</i>	12	−0.012	−0.060	0.008	−0.260	−0.269	−0.768	0.178	−0.021	−0.007	−0.062	0.032	1	
<i>FirmInnovationEfficiency</i>	13	0.117	0.054	0.075	−0.024	0.033	−0.089	0.100	−0.088	−0.027	−0.012	0.057	0.090	1

Appendix A4. Correlation matrix for the firm-level woman-led invention network

	1	2	3	4	5	6	7	8	9	10	11	12	13
<i>FirmPat_t</i>	1	1											
<i>FirmPatCit_t</i>	2	0.683	1										
<i>FemFirmBetweenness_t</i>	3	0.209	0.235	1									
<i>FemFirmClustering_t</i>	4	0.007	0.032	0.113	1								
<i>FemFirmTriangles_t</i>	5	0.044	0.095	0.298	0.779	1							

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		1	2	3	4	5	6	7	8	9	10	11	12	13
<i>PatInventorNum_{it}</i>	6	−0.003	0.063	0.121	0.202	0.221	1							
<i>PatActivityTime_{it}</i>	7	0.204	0.188	0.300	−0.031	0.052	−0.040	1						
<i>PatWithdrawal_{it}</i>	8	−0.084	−0.095	−0.086	0.018	0.005	−0.013	−0.206	1					
<i>PatRefusal_{it}</i>	9	−0.020	−0.019	−0.032	−0.009	−0.022	0.013	−0.027	−0.107	1				
<i>PatCountryShare_{it}</i>	10	−0.019	0.018	0.076	0.189	0.176	0.105	−0.089	0.066	−0.013	1			
<i>FemFirmGlobalCohesion_{it}</i>	11	0.018	−0.044	0.005	−0.054	−0.123	−0.002	0.179	−0.145	−0.005	−0.045	1		
<i>PatInventorShare</i>	12	0.0641	−0.009	−0.041	−0.185	−0.176	−0.810	0.179	−0.042	−0.024	−0.131	0.016	1	
<i>FirmInnovationEfficiency</i>	13	0.676	0.450	0.191	0.005	0.057	−0.11	0.162	−0.169	−0.049	−0.025	0.046	0.175	1

Appendix A5. – Regression formulas

Man-led network:

$$\begin{aligned} \left[\begin{array}{c} \text{FirmPat}_{it} \\ \text{FirmPatCit}_{it} \end{array} \right] &= \alpha + \beta_1 \ln(10^8 \times \text{FirmBetweenness}_{it} + 1) + \beta_2 \ln(\text{PatInventorShare}_{it} + 1) + \beta_3 \ln(\text{PatActivityTime}_{it} + 1) \\ &\quad + \beta_4 \ln(\text{PatRefusal}_{it} + 1) + \beta_5 \ln(\text{PatWithdrawal}_{it} + 1) + \beta_6 \ln(\text{PatCountryShare}_{it} + 1) \\ &\quad + \beta_7 \text{FirmGlobalCohesion}_{it} + \beta_8 \text{PatInventorNum}_{it} + \beta_9 \text{FirmInnovationEfficiency}_{it} + \beta_{10} \text{Year} + \delta_i + \varepsilon_i \\ \left[\begin{array}{c} \text{FirmPat}_{it} \\ \text{FirmPatCit}_{it} \end{array} \right] &= \alpha + \beta_1 \ln(10^8 \times \text{FirmTriangle}_{it} + 1) + \beta_2 \ln(\text{PatInventorShare}_{it} + 1) + \beta_3 \ln(\text{PatActivityTime}_{it} + 1) \\ &\quad + \beta_4 \ln(\text{PatRefusal}_{it} + 1) + \beta_5 \ln(\text{PatWithdrawal}_{it} + 1) + \beta_6 \ln(\text{PatCountryShare}_{it} + 1) \\ &\quad + \beta_7 \text{FirmGlobalCohesion}_{it} + \beta_8 \text{PatInventorNum}_{it} + \beta_9 \text{FirmInnovationEfficiency}_{it} + \beta_{10} \text{Year} + \delta_i + \varepsilon_i \\ \left[\begin{array}{c} \text{FirmPat}_{it} \\ \text{FirmPatCit}_{it} \end{array} \right] &= \alpha + \beta_1 \ln(10^8 \times \text{FirmClustering}_{it} + 1) + \beta_2 \ln(\text{PatInventorShare}_{it} + 1) + \beta_3 \ln(\text{PatActivityTime}_{it} + 1) \\ &\quad + \beta_4 \ln(\text{PatRefusal}_{it} + 1) + \beta_5 \ln(\text{PatWithdrawal}_{it} + 1) + \beta_6 \ln(\text{PatCountryShare}_{it} + 1) \\ &\quad + \beta_7 \text{PatInventorNum}_{it} + \beta_8 \text{FirmInnovationEfficiency}_{it} + \beta_9 (\text{FirmGlobalCohesion}_{it} \times \text{FirmClustering}_{it}) + \beta_{10} \text{Year} + \delta_i + \varepsilon_i \end{aligned}$$

Woman-led network:

$$\begin{aligned} \left[\begin{array}{c} \text{FemFirmPat}_{it} \\ \text{FemFirmPatCit}_{it} \end{array} \right] &= \alpha + \beta_1 \ln(10^8 \times \text{FemFirmBetweenness}_{it} + 1) + \beta_2 \ln(\text{PatInventorShare}_{it} + 1) + \beta_3 \ln(\text{PatActivityTime}_{it} + 1) \\ &\quad + \beta_4 \ln(\text{PatRefusal}_{it} + 1) + \beta_5 \ln(\text{PatWithdrawal}_{it} + 1) + \beta_6 \ln(\text{PatCountryShare}_{it} + 1) \\ &\quad + \beta_7 \text{FemFirmGlobalCohesion}_{it} + \beta_8 \text{PatInventorNum}_{it} + \beta_9 \text{FirmInnovationEfficiency}_{it} + \beta_{10} \text{Year} + \delta_i + \varepsilon_i \\ \left[\begin{array}{c} \text{FemFirmPat}_{it} \\ \text{FemFirmPatCit}_{it} \end{array} \right] &= \alpha + \beta_1 \ln(10^8 \times \text{FemFirmTriangle}_{it} + 1) + \beta_2 \ln(\text{PatInventorShare}_{it} + 1) + \beta_3 \ln(\text{PatActivityTime}_{it} + 1) \\ &\quad + \beta_4 \ln(\text{PatRefusal}_{it} + 1) + \beta_5 \ln(\text{PatWithdrawal}_{it} + 1) + \beta_6 \ln(\text{PatCountryShare}_{it} + 1) \\ &\quad + \beta_7 \text{FemFirmGlobalCohesion}_{it} + \beta_8 \text{PatInventorNum}_{it} + \beta_9 \text{FirmInnovationEfficiency}_{it} + \beta_{10} \text{Year} + \delta_i + \varepsilon_i \\ \left[\begin{array}{c} \text{FemFirmPat}_{it} \\ \text{FemFirmPatCit}_{it} \end{array} \right] &= \alpha + \beta_1 \ln(10^8 \times \text{FemFirmClustering}_{it} + 1) + \beta_2 \ln(\text{PatInventorShare}_{it} + 1) + \beta_3 \ln(\text{PatActivityTime}_{it} + 1) \\ &\quad + \beta_4 \ln(\text{PatRefusal}_{it} + 1) + \beta_5 \ln(\text{PatWithdrawal}_{it} + 1) + \beta_6 \ln(\text{PatCountryShare}_{it} + 1) \\ &\quad + \beta_7 \text{PatInventorNum}_{it} + \beta_8 \text{FirmInnovationEfficiency}_{it} + \beta_9 (\text{FemFirmGlobalCohesion}_{it} \times \text{FemFirmClustering}_{it}) \\ &\quad + \beta_{10} \text{Year} + \delta_i + \varepsilon_i \end{aligned}$$

Appendix A6. Tests of difference in coefficients

We used two statistical models to test whether the effect of betweenness centrality differ between man-led and woman-led networks for our two dependent variables **FirmPat** and **FirmPatCit**.

1. The Wald Test (`test` command) checks if the coefficient for our variable is statistically different across the two groups. It evaluates the null hypothesis that the coefficients are equal.
2. The Linear Combination Test (`lincom` command) calculates the actual difference between the coefficients, along with its standard error, z-score, and confidence interval, providing a more precise estimate of the difference.

Number of patents (*FirmPat*)

Test Results for Differences in Coefficients in the Number of Patents Models.

Independent Variable	Coefficient Difference	Standard Error	z-score	p-value	95 % Confidence Interval
<i>FirmBetweenness</i>	0.0076	0.0045	1.70	0.0897	[-0.0012, 0.0163]

Chi-Square Tests for Coefficient Differences.

- *FirmBetweenness*: $\chi^2(1) = 2.88$, $p = 0.0897$
- ***FirmBetweenness*** is statistically significant at the 10 % level ($p = 0.090$), indicating a moderate difference between man- and woman-led networks.
- The coefficient difference for *FirmBetweenness* is 0.0076, indicating that its effect on the number of patents is greater in man-led networks than in woman-led networks.
- The one-sided test (H1: difference >0) has a p-value of 0.0448, which is statistically significant at the 5 % level, supporting the claim that *FirmBetweenness* has a stronger impact on patent output in man-led networks.
- The one-sided test (H1: difference <0) has a p-value of 0.9552, meaning there is no support for the hypothesis that *FirmBetweenness* has a stronger effect in woman-led networks.

Citations of Patents: (*FirmPatCit*)

Test Results for Differences in Coefficients in the Number of citation Models

Independent Variable	Coefficient Difference	Standard Error	z-score	p-value	95 % Confidence Interval
<i>FirmBetweenness</i>	0.0880	0.0191	4.62	0.0000	[0.0507, 0.1254]

The chi-squared test results:

- *FirmBetweenness*: $\chi^2(1) = 21.32$, $p = 0.0000$

FirmBetweenness is highly statistically significant at the 1 % level ($p = 0.0000$), indicating that betweenness centrality has a significantly stronger effect on citation counts in man-led networks.

- The coefficient difference is 0.0880, suggesting that *FirmBetweenness* contributes much more to citation impact in man-led networks than in woman-led networks.
- The one-sided test (H1: difference >0) has a p-value of 0.000002, which is highly significant, confirming this stronger effect in man-led networks.

Appendix A7. –Robustness check

Table A7-1

The regression results of the impact of Betweenness centrality and Clustering on the number of patents in man-led network

Variables	Models					
FirmPat _t	Model 1		Model 2		Model 3	
Network variables						
FirmBetweenness _t	0.031	***				
	(0.002)					
FirmTriangles _t			−0.199	***		
			(0.013)			
FirmClustering _t					−0.091	***
					(0.013)	
Control variables						
PatInventorShare _t	0.338	***	0.341	***	0.346	***
	(0.013)		(0.014)		(0.014)	
PatActivityTime _t	0.356	***	0.371	***	0.423	***
	(0.011)		(0.012)		(0.012)	
PatWithdrawal _t	−0.507	***	−0.506	***	−0.517	***
	(0.021)		(0.021)		(0.023)	
PatRefusal _t	−0.156	***	−0.155	***	−0.156	***

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Table A7-1 (continued)

Variables	Models		
<i>FirmPat_t</i>	Model 1	Model 2	Model 3
	(0.012)	(0.012)	(0.012)
<i>PatCountryShare_t</i>	−0.048 ***	−0.056 ***	−0.040 ***
	(0.012)	(0.011)	(0.012)
<i>PatInventorNum_t</i>	0.778 ***	0.787 ***	0.821 ***
	(0.014)	(0.015)	(0.015)
<i>FirmGlobalCohesion_t</i>	−1.047 ***	−4.379 ***	
	(0.103)	(0.248)	
<i>FirmInnovationEfficiency</i>	0.612 ***	0.605 ***	0.678 ***
	(0.041)	(0.041)	(0.043)
<i>Dummy_year</i>	Yes	Yes	Yes
<i>Constant</i>	10.670 ***	50.750 ***	0.509 ***
	(1.010)	(2.851)	(0.072)
<i>lnalpha_constant</i>	−0.670 ***	−0.674 ***	−0.622 ***
	(0.027)	(0.027)	(0.029)
<i>Nb of observations</i>	23752	23752	23752
<i>Nb of groups</i>	7698	7698	7698
<i>R²_pseudo</i>	0.2331	0.2336	0.2282
<i>Loglikelihood</i>	−65536	−65491	−65956

Note: ***, ** and * show significance at the 1 %, 5 % and 10 % levels, respectively, and standard errors are presented in parentheses.

Table A7-2

The regression results of the impact of Betweenness centrality and Clustering on the number of patent citations in man-led network

Variables	Models		
<i>FirmPatCit_t</i>	Model 1	Model 2	Model 3
Network variables			
<i>FirmBetweenness_t</i>	0.030 ***		
	(0.005)		
<i>FirmTriangles_t</i>		−0.098 ***	
		(0.018)	
<i>FirmClustering_t</i>			−0.093 **
			(0.044)
Control variables			
<i>PatInventorShare_t</i>	−0.091 **	−0.088 *	−0.076 *
	(0.044)	(0.046)	(0.045)
<i>PatActivityTime_t</i>	0.468 ***	0.462 ***	0.557 ***
	(0.043)	(0.041)	(0.042)
<i>PatWithdrawal_t</i>	−0.380 ***	−0.395 ***	−0.381 ***
	(0.053)	(0.052)	(0.055)
<i>PatRefusal_t</i>	−0.133 *	−0.142 **	−0.133 *
	(0.071)	(0.072)	(0.072)
<i>PatCountryShare_t</i>	0.156 ***	0.154 ***	0.170 ***
	(0.040)	(0.040)	(0.043)
<i>PatInventorNum_t</i>	0.515 ***	0.531 ***	0.552 ***
	(0.048)	(0.049)	(0.048)
<i>FirmGlobalCohesion_t</i>	0.324		
	(0.325)		
<i>FirmInnovationEfficiency</i>	0.606 ***	0.655 ***	0.658 ***
	(0.041)	(0.045)	(0.043)
<i>Dummy_year</i>	Yes	Yes	Yes
<i>Constant</i>	−4.109	1.040 ***	−0.952 ***
	(3.092)	(0.045)	(0.202)
<i>lnalpha_constant</i>	1.432 ***	1.460 ***	1.440 ***
	(0.028)	(0.026)	(0.027)
<i>Nb of observations</i>	23752	23752	23752
<i>Nb of groups</i>	7698	7698	7698
<i>R²_pseudo</i>	0.0966	0.0935	0.0956
<i>Loglikelihood</i>	−26480	−26572	−26509

Note: ***, ** and * show significance at the 1 %, 5 % and 10 % levels, respectively, and standard errors are presented in parentheses.

Table A7-3

The regression results of the impact of Betweenness centrality and Clustering on the number of patents in woman-led network

Variables	Models		
<i>FirmPat_t</i>	Model 1	Model 2	Model 3
Network variables			
<i>FemFirmBetweenness_t</i>	0.016 (0.003)		
<i>FemFirmTriangles_t</i>		0.139 (0.018)	
<i>FemFirmClustering_t</i>			0.178 (0.018)
Control variables			
<i>PatInventorShare_t</i>	0.322 (0.026)	0.317 (0.027)	0.323 (0.026)
<i>PatActivityTime_t</i>	0.337 (0.019)	0.329 (0.019)	0.361 (0.018)
<i>PatWithdrawal_t</i>	−0.470 (0.020)	−0.463 (0.020)	−0.470 (0.020)
<i>PatRefusal_t</i>	−0.142 (0.014)	−0.137 (0.014)	−0.140 (0.014)
<i>PatCountryShare_t</i>	−0.061 (0.011)	−0.059 (0.011)	−0.052 (0.011)
<i>PatInventorNum_t</i>	0.629 (0.024)	0.629 (0.024)	0.646 (0.024)
<i>FemFirmGlobalCohesion_t</i>	1.354 (0.211)	3.908 (0.363)	
<i>FirmInnovationEfficiency</i>	1.045 (0.063)	1.002 (0.063)	1.103 (0.062)
<i>Dummy_year</i>	Yes	Yes	Yes
Constant	14.346 (2.019)	44.864 (4.172)	0.784 (0.092)
<i>lnalpha_constant</i>	−0.846 (0.036)	−0.866 (0.036)	−0.831 (0.037)
<i>Nb of observations</i>	6325	6325	6325
<i>Nb of groups</i>	4043	4043	4043
<i>R²_pseudo</i>	0.1671	0.1695	0.1659
<i>Loglikelihood</i>	−16510	−16461	−16533

Note: ***, ** and * show significance at the 1 %, 5 % and 10 % levels, respectively, and standard errors are presented in parentheses.

Table A7-4

– The regression results of the impact of Betweenness centrality and Clustering on the number of citations in woman-led network

Variables	Models		
<i>FirmPat_tCit</i>	Model 1	Model 2	Model 3
Network variables			
<i>FemFirmBetweenness_t</i>	0.009 (0.006)		
<i>FemFirmTriangles_t</i>		0.013 (0.019)	
<i>FemFirmClustering_t</i>			0.156 (0.053)
Control variables			
<i>PatInventorShare_t</i>	−0.028 (0.059)	−0.032 (0.059)	−0.105 (0.077)
<i>PatActivityTime_t</i>	0.277 (0.053)	0.293 (0.052)	0.501 (0.071)
<i>PatWithdrawal_t</i>	−0.361 (0.058)	−0.368 (0.059)	−0.240 (0.072)
<i>PatRefusal_t</i>	−0.069 (0.049)	−0.071 (0.050)	−0.057 (0.059)
<i>PatCountryShare_t</i>	0.071 (0.022)	0.070 (0.023)	0.064 (0.035)
<i>PatInventorNum_t</i>	0.544 (0.051)	0.537 (0.052)	0.469 (0.067)
<i>FemFirmGlobalCohesion_t</i>	0.241 (0.043)		
<i>FirmInnovationEfficiency</i>	1.445 (0.112)	1.472 (0.112)	1.241 (0.106)
<i>Dummy_year</i>	Yes	Yes	Yes

(continued on next page)

Table A7-4 (continued)

Variables	Models					
Constant	1.199 (0.090)	***	−1.073 (0.454)	**	−1.161 (0.472)	**
<i>lnalpha_constant</i>	1.531 (0.029)	***	1.532 (0.029)	***	1.471 (0.040)	***
<i>Nb of observations</i>	6325		6325		6325	
<i>Nb of groups</i>	4043		4043		4043	
<i>R²_pseudo</i>	0.0533		0.0532		0.0623	
<i>Loglikelihood</i>	−13191		−13193		−7344	

Note: ***, ** and * show significance at the 1 %, 5 % and 10 % levels, respectively, and standard errors are presented in parentheses.

Data availability

Data will be made available on request.

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